

Graduation Project

GAN GOSH

SIC 603

SAMSUNG

Graduation Project

Together for Tomorrow!
Enabling People
Education for Future Generations

Team Members



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Lenda Deif



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Image Style transfer



Deep Learning Project

Project's Agenda

- | **Problem Description**

- | EDA

- | Project Workflow

- | Pre Processing

- | Modeling

- | Post Processing

- | Testing & Deployment

- | Conclusions & Future Improvement

Problem Description



Targeted Audience

- **Art Lovers**
- **Art students**
- **Artists around the globe**

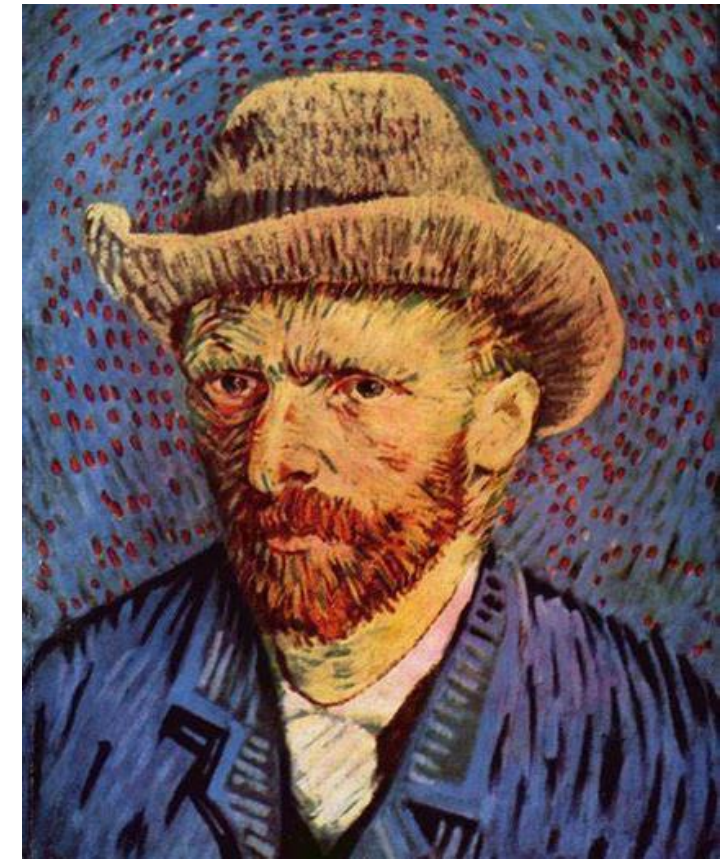


Problem Description

- Ever wondered how a certain artist will draw a certain photo if they were alive?
- Ever wondered how a collab between Monet and Van Gogh will look like?



Art is an Identity ✨

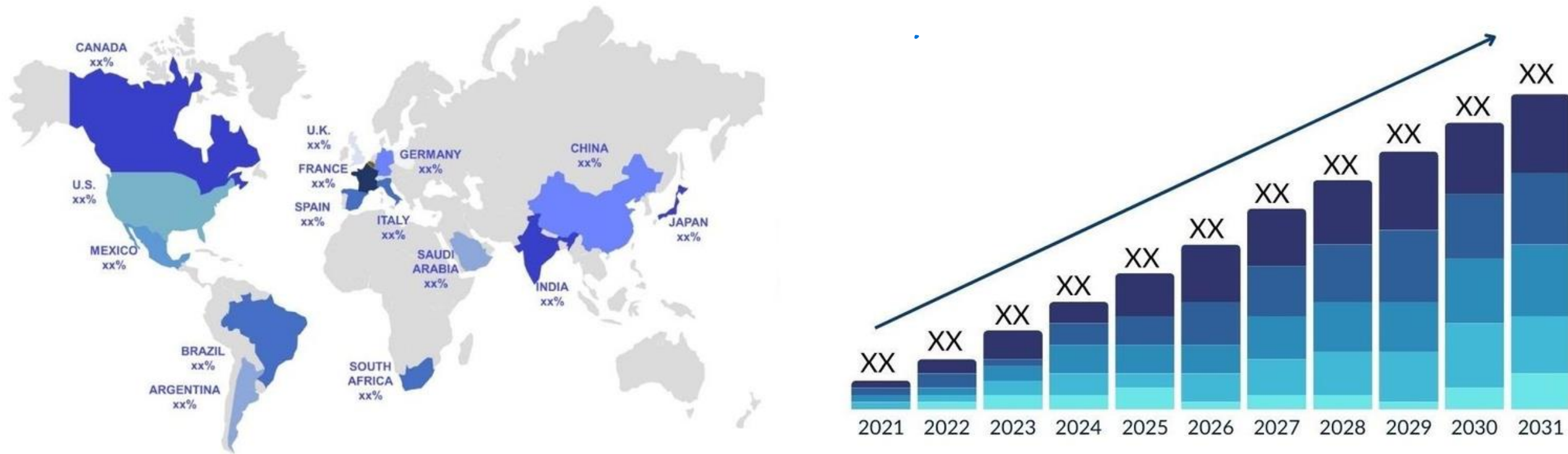


Statistics around the problem

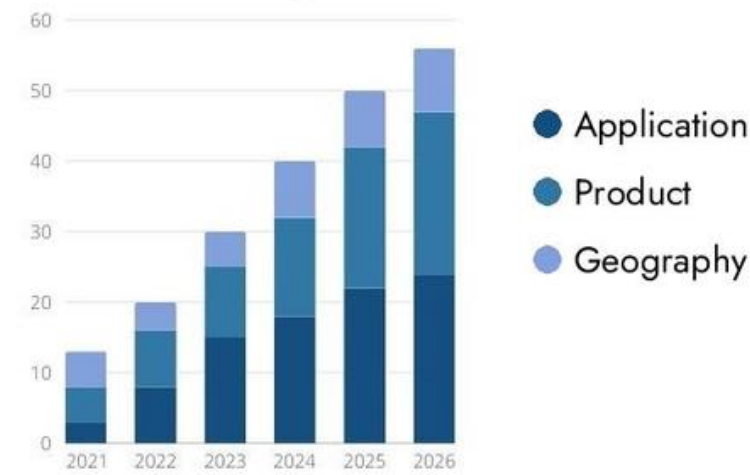
- **Popularity of Artistic Transformations:** 60-70% of users prefer enhancing photos with artistic effects.
- **Growth of Digital Art Apps:** The market for AI-driven apps like Prisma is growing at 20% annually.
- **Increased Social Media Engagement:** Artistic images boost engagement by 30% on social media, encouraging sharing.

Statistics around the problem

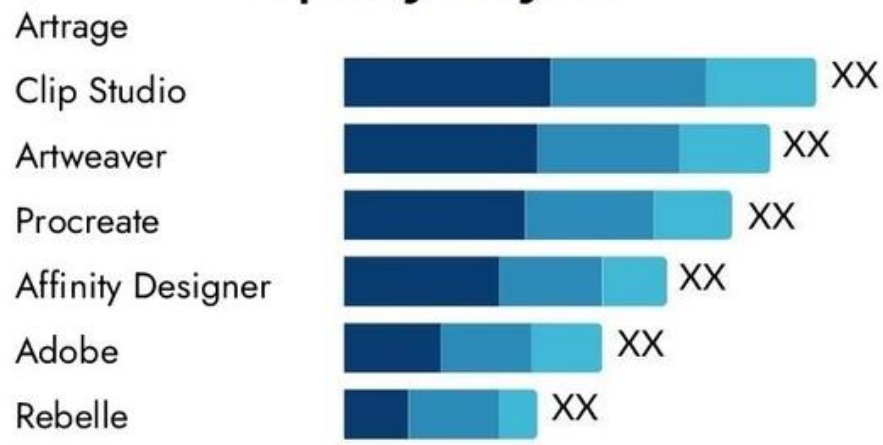
Digital Art Software Market Size and Scope



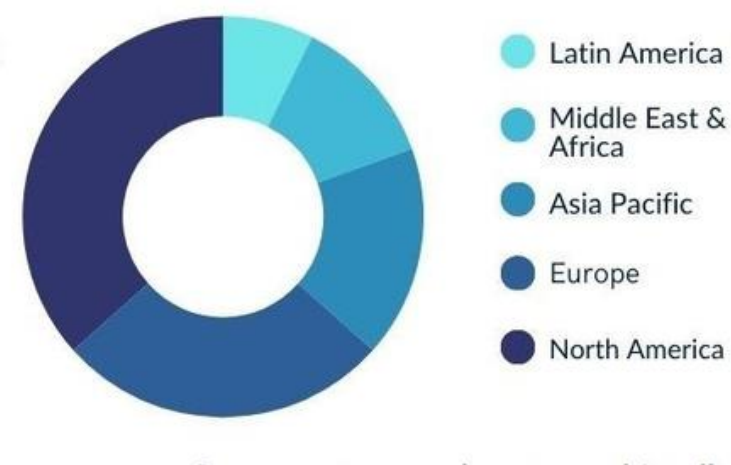
Market Segmentation



Top Key Players



Regional Analysis



Academic Value & Applications



Digital Production



Virtual Exhibitions

Project Aim



**Style
transfer**



Project's Agenda

| Problem Description

| **EDA**

| Project Workflow

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Some styling samples 'Monet Art'



Some Actual photos to apply style to it.



Project's Agenda

| Problem Description

| EDA

| **Project Workflow**

| Pre Processing

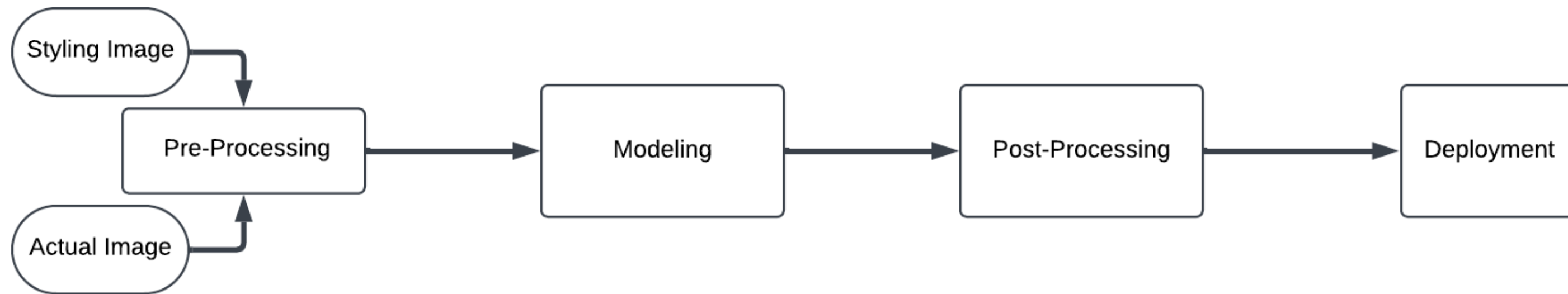
| Modeling

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Workflow



Every phase will have it's own pipeline.

Project's Agenda

| Problem Description

| EDA

| Project Workflow

| **Pre Processing**

| Modeling

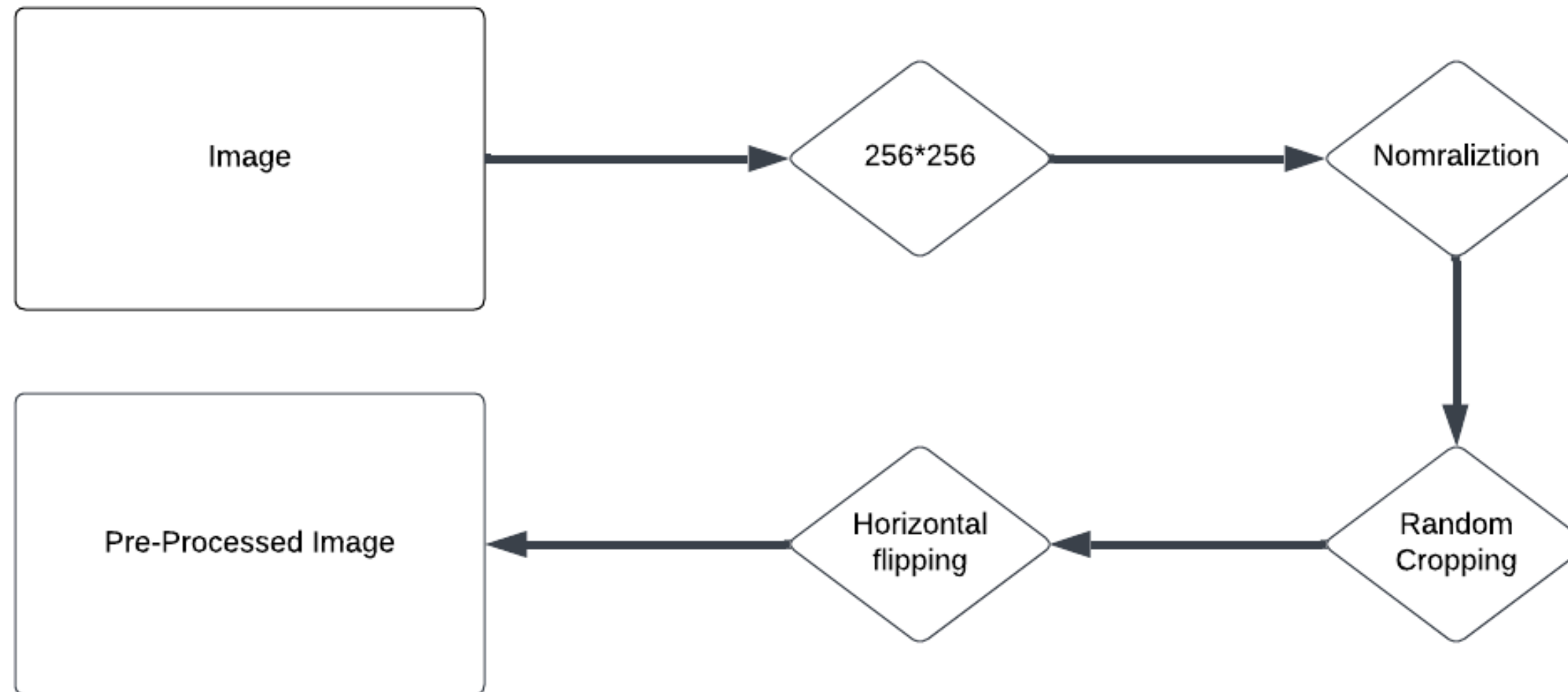
| Post Processing

| Testing & Deployment

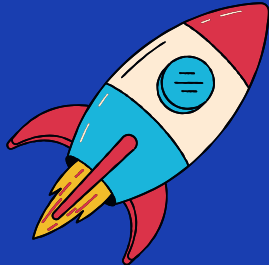
| Conclusions & Future Improvement

Pre-Processing

We decided to follow the approach mentioned in the model's paper.

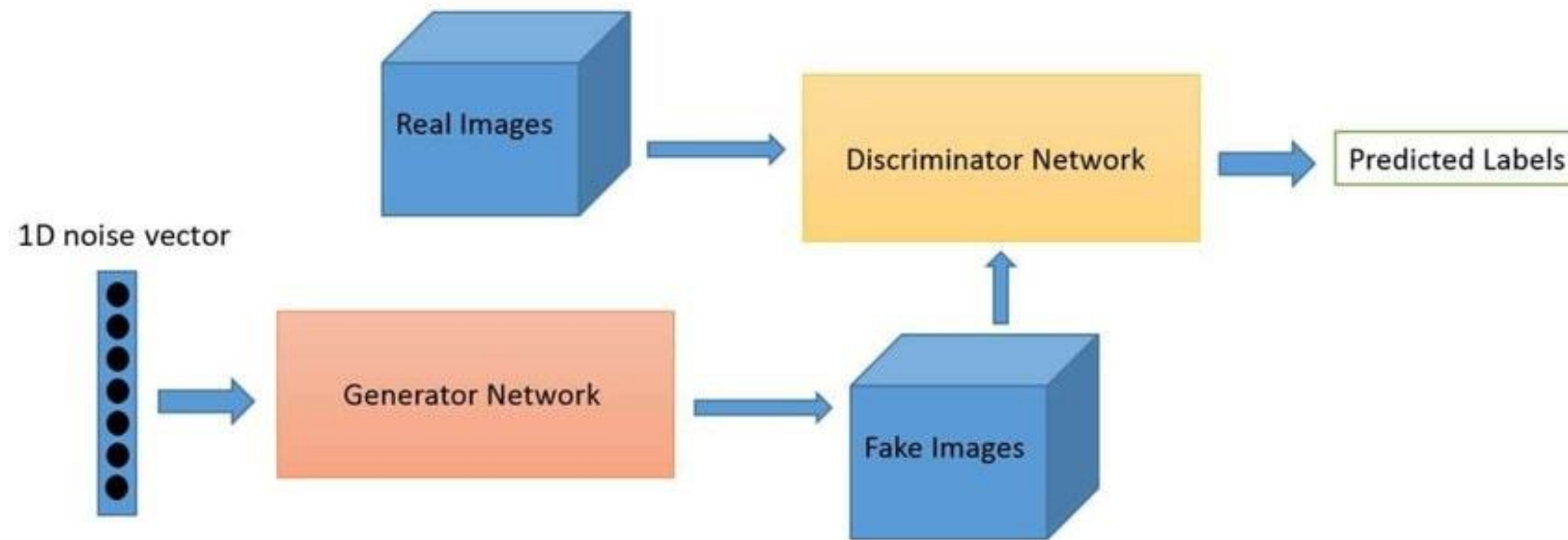


Now with most interesting part,

Modeling 

Modeling

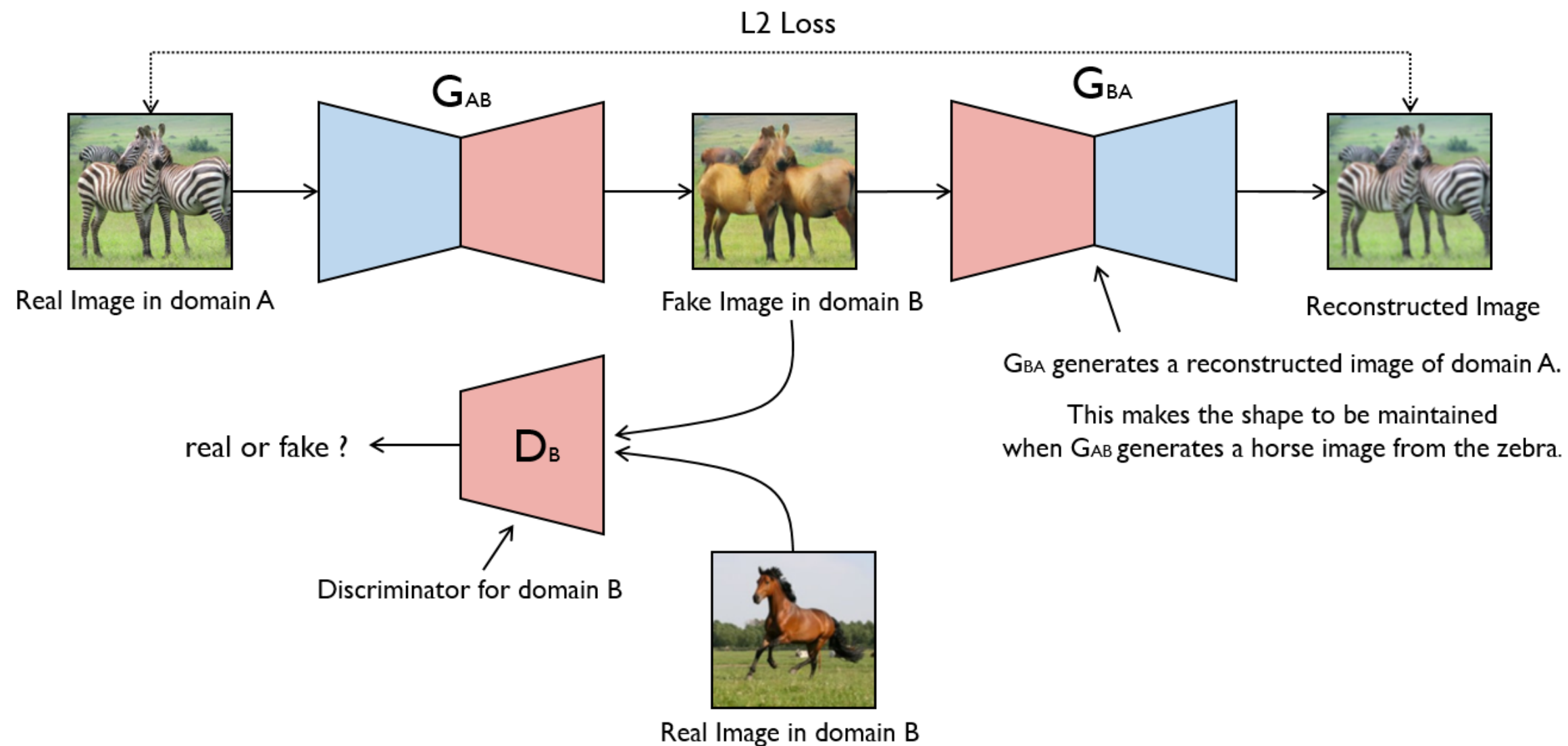
Firstly, what's GANs ?



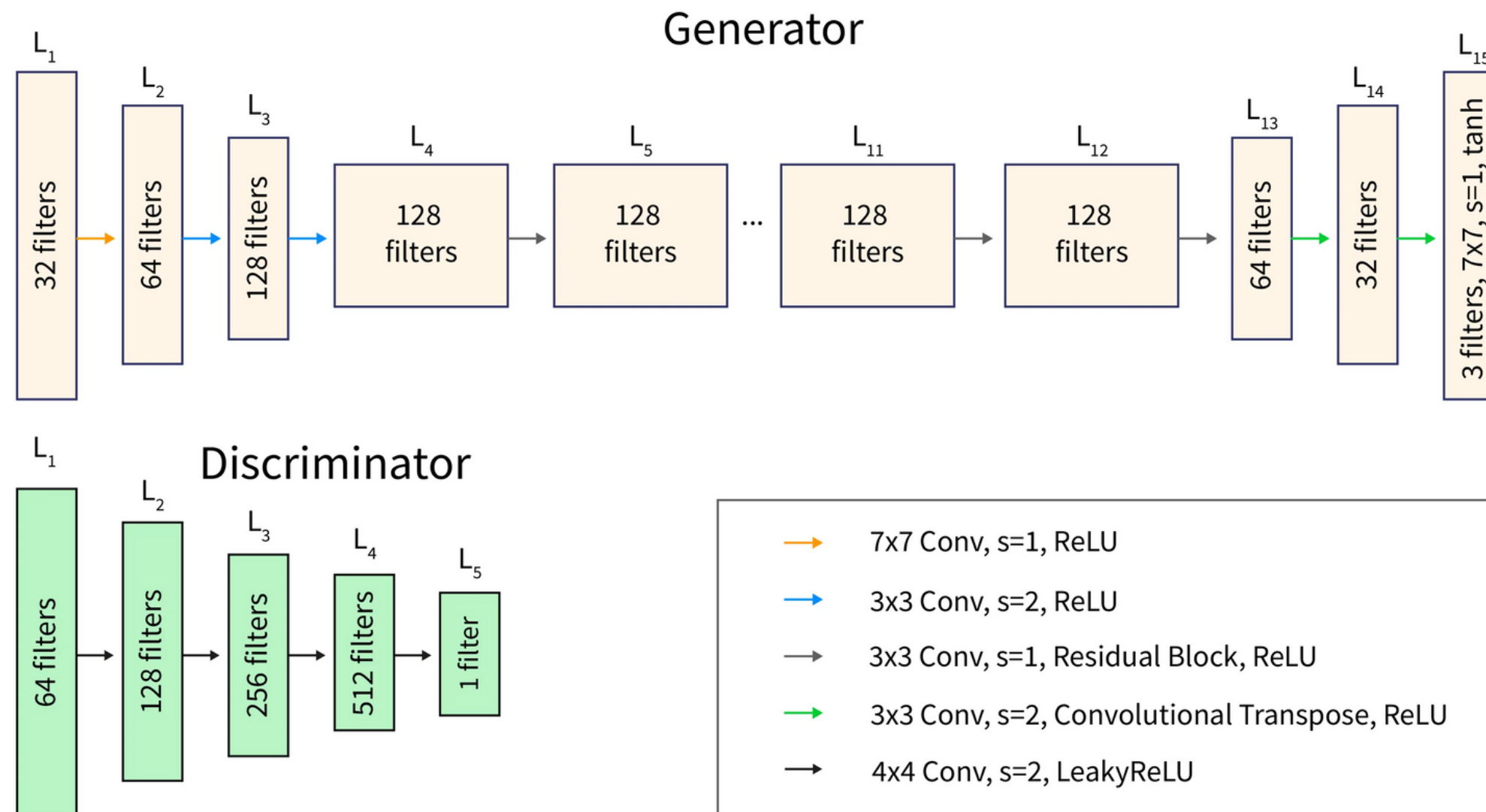
There're many architectures that follows this concept and we decided to use one of them, which's cycleGAN

Modeling

We used the **CycleGAN** architecture , which's used for style transferring and generating new images generally.



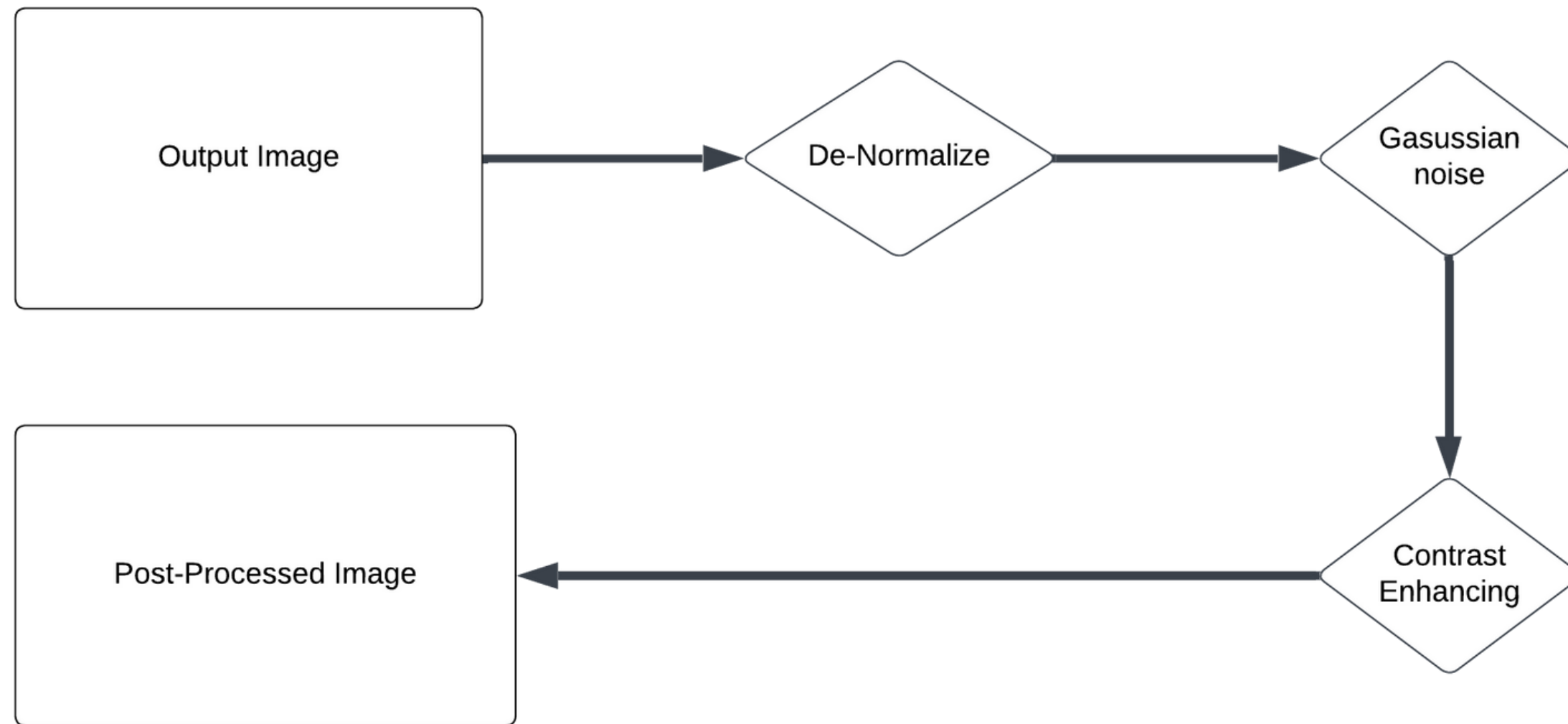
Modeling



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- | Modeling
- | Post Processing**
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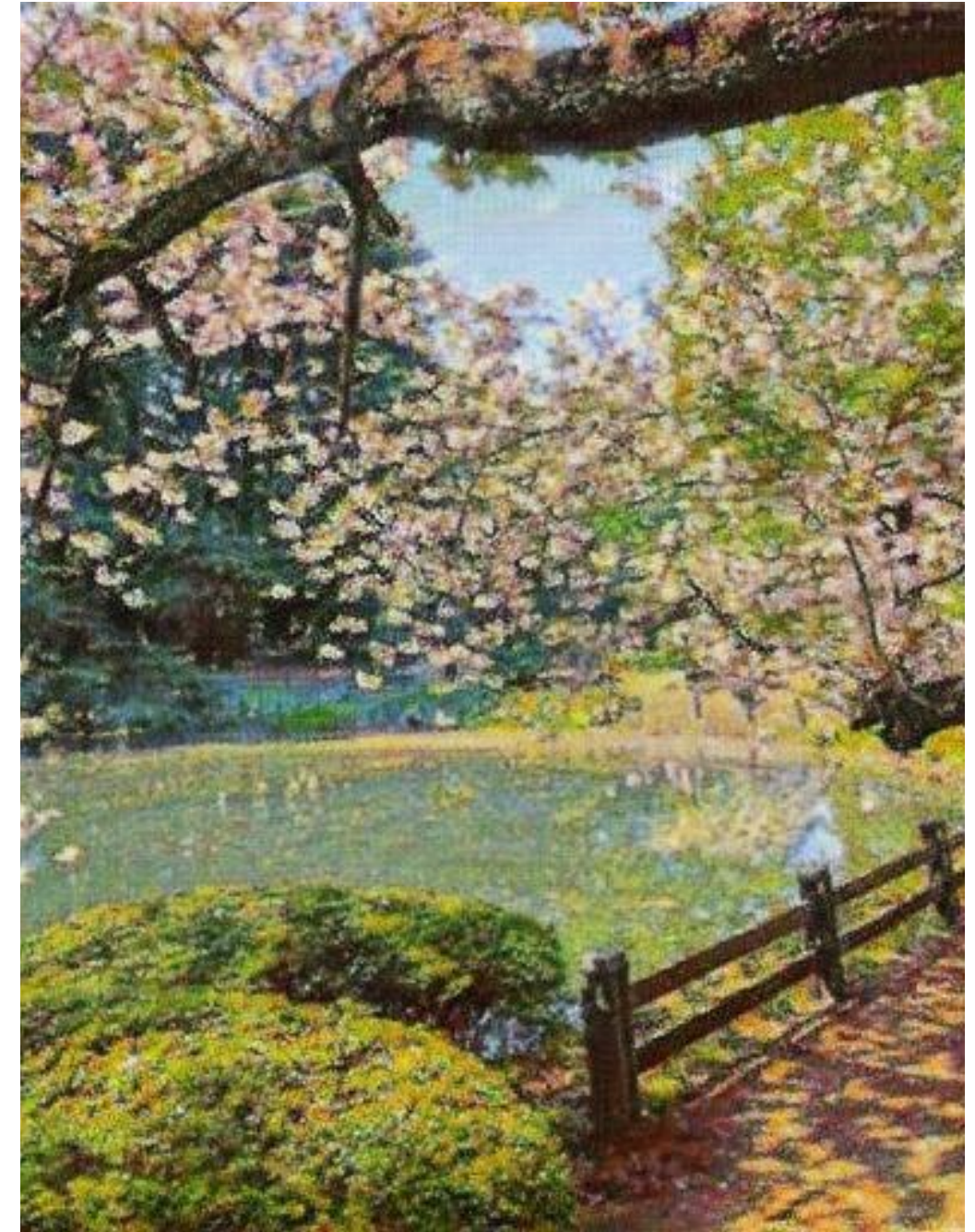
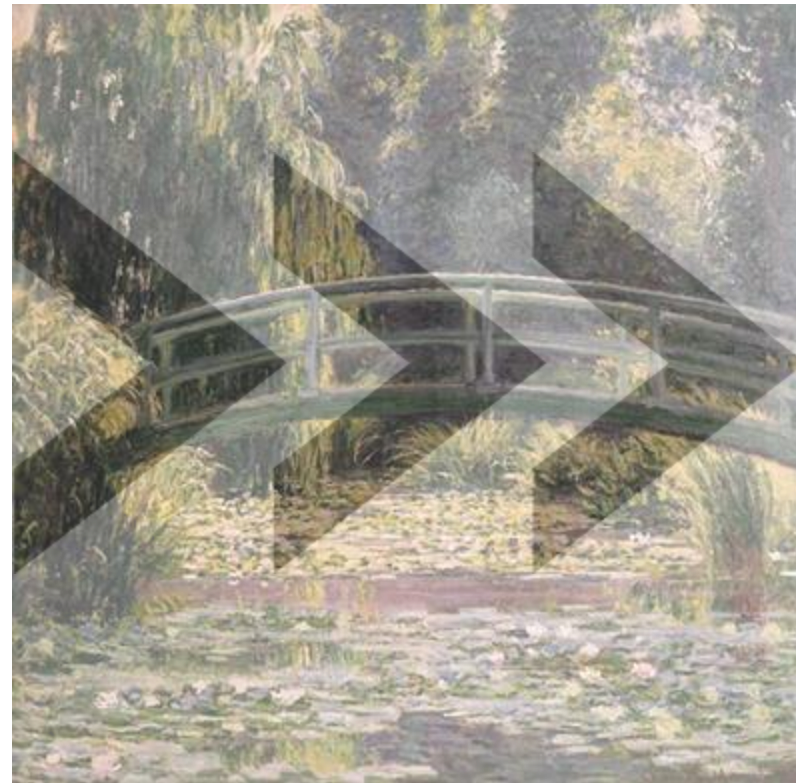
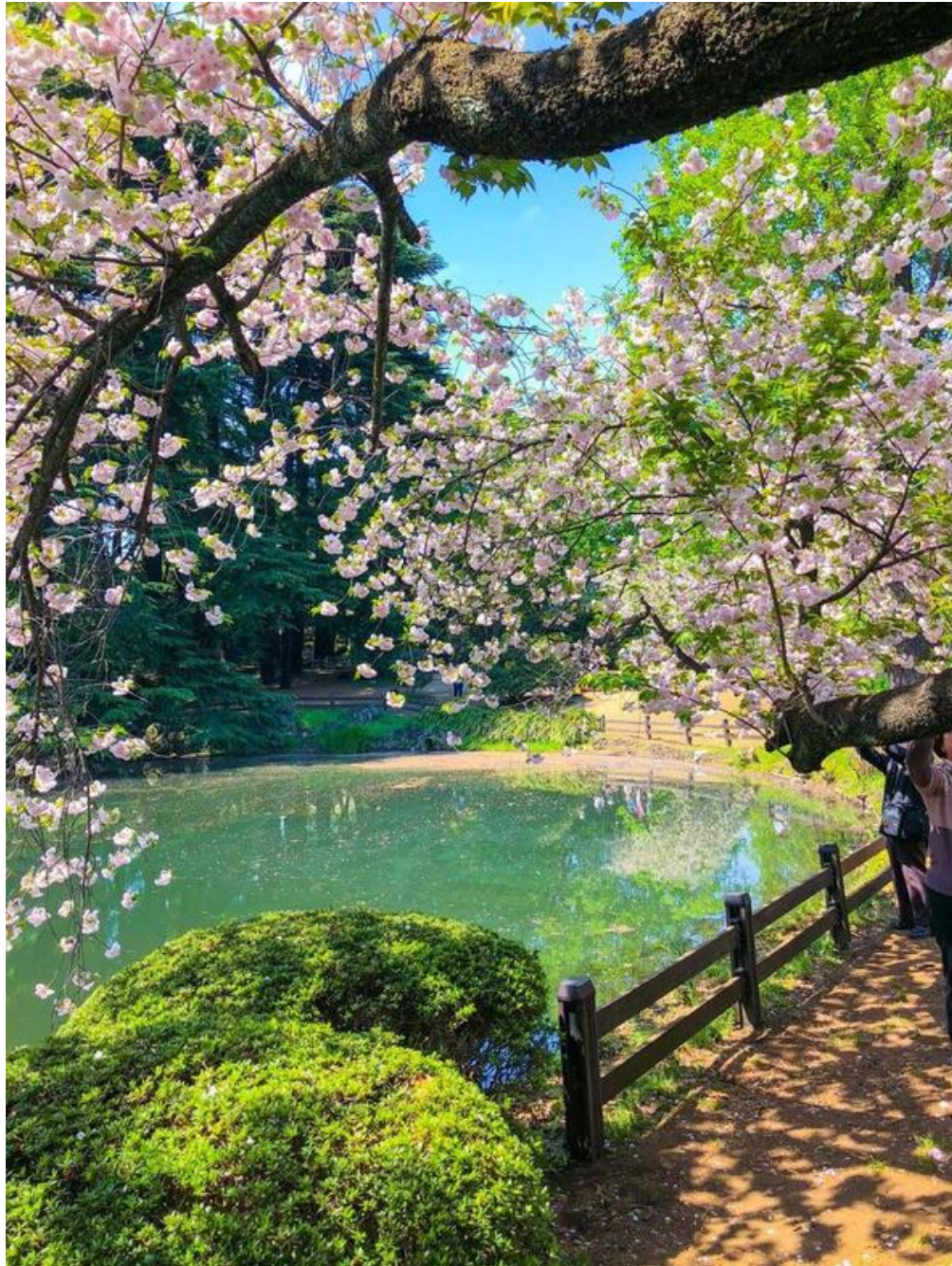
Post-Processing



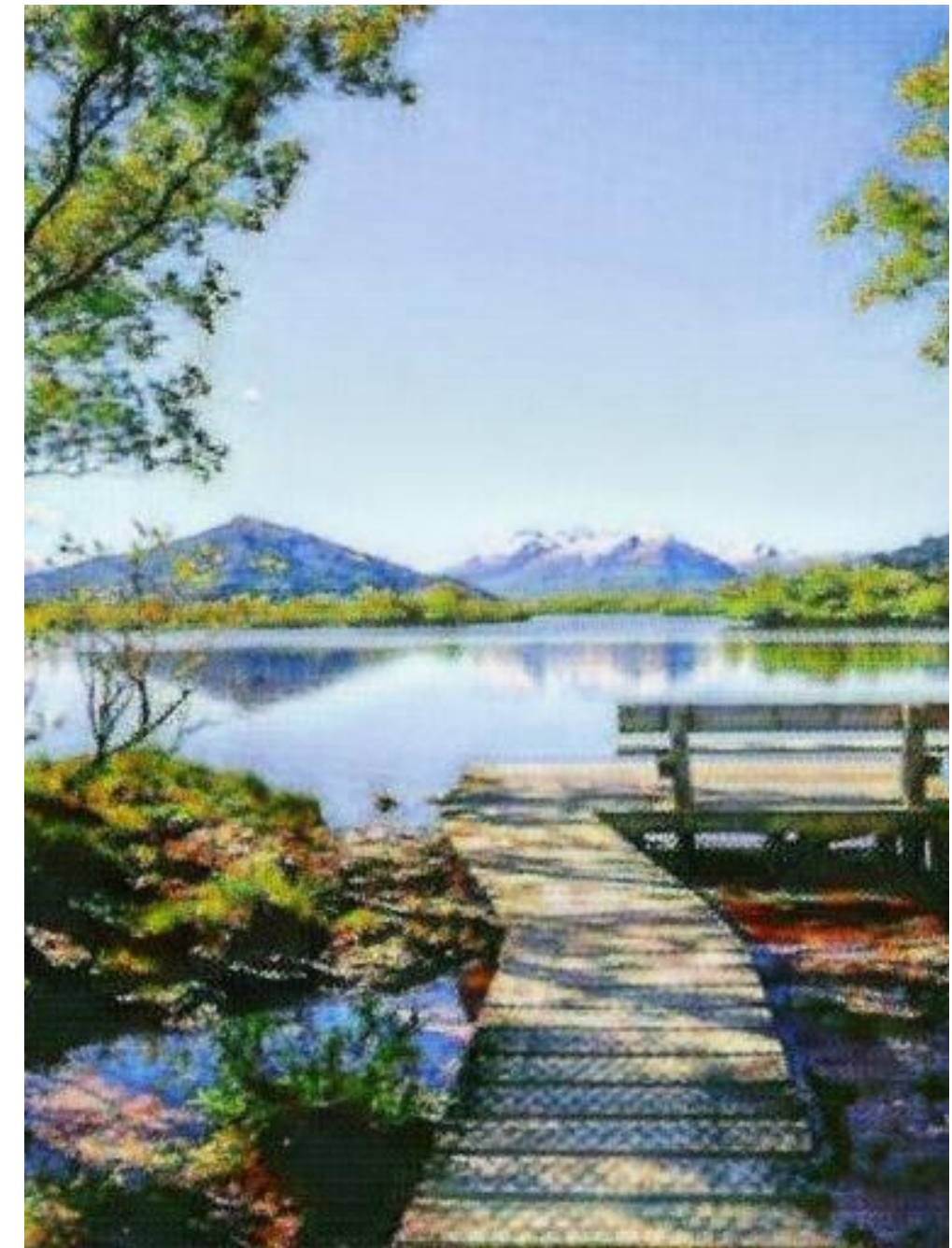
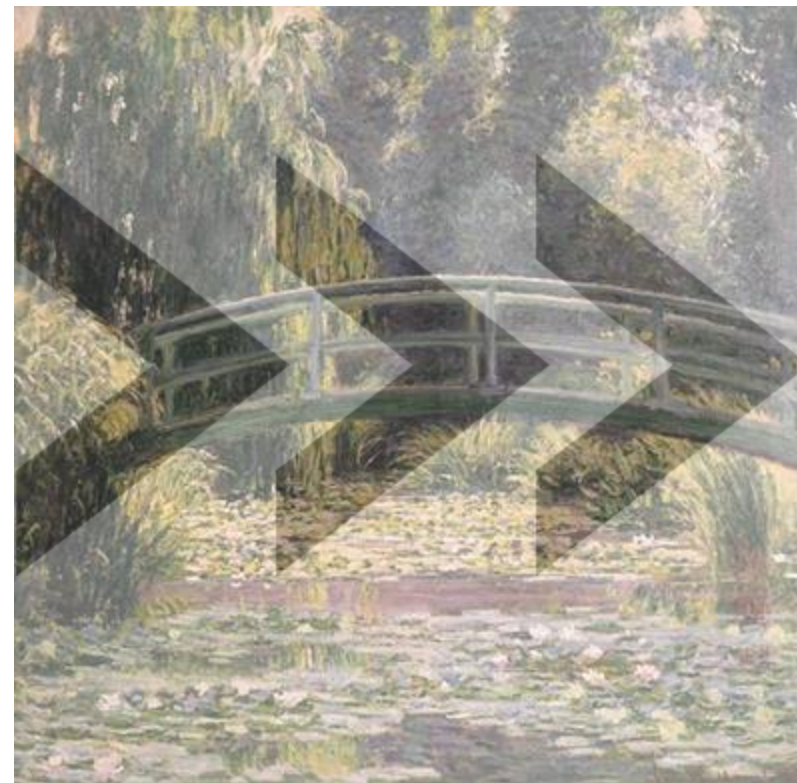
Project's Agenda

- | Problem Description
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- | Modeling
- | Post Processing
- | Results & Deployment**
- | Conclusions & Future Improvement

Results

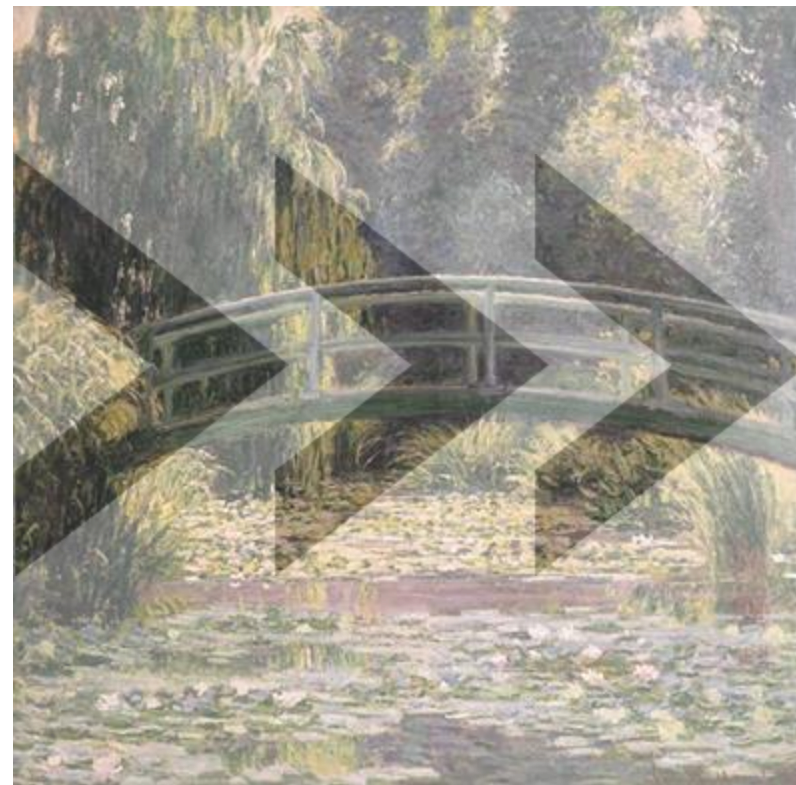


Results



Limited only to photos?

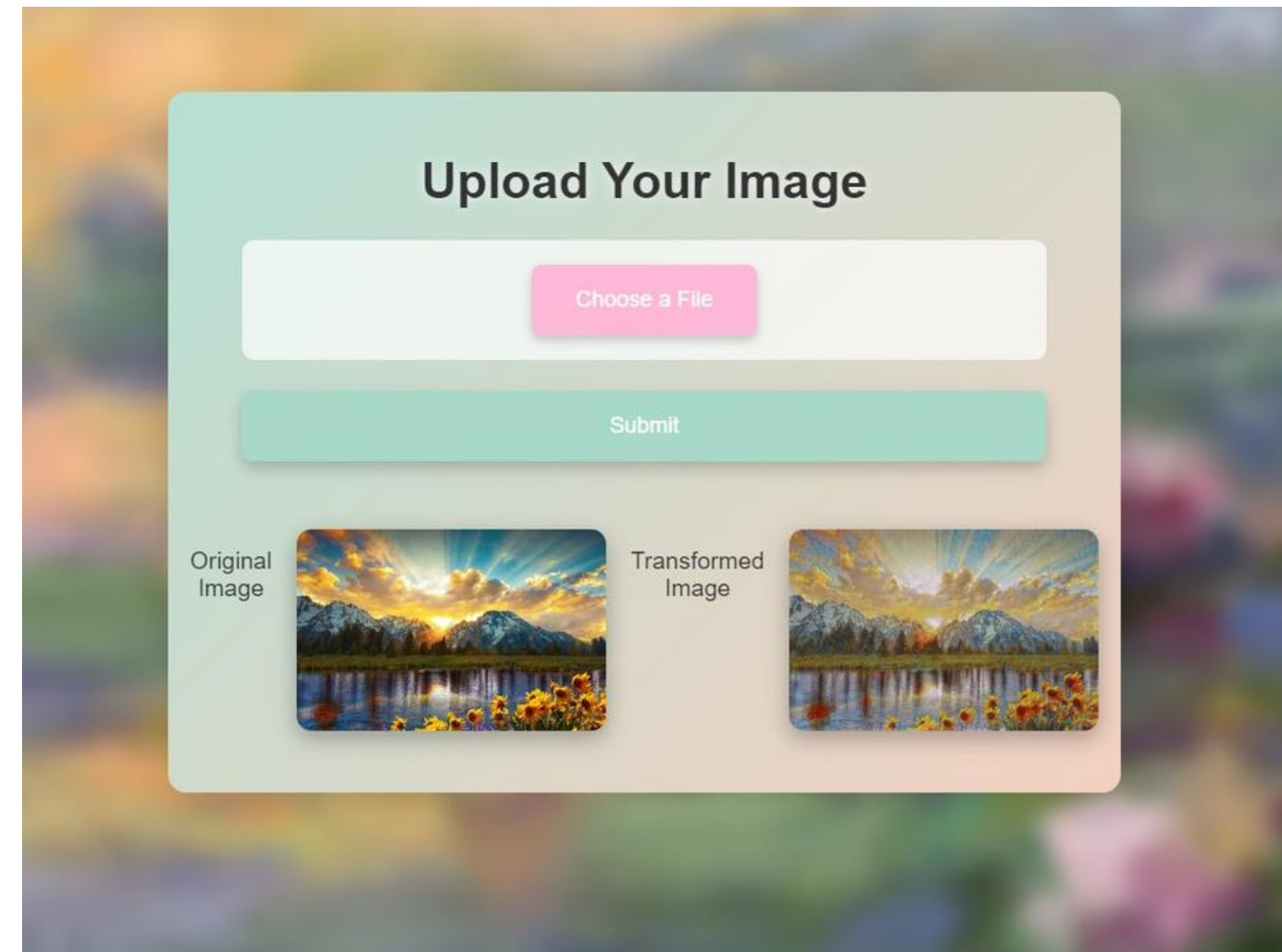
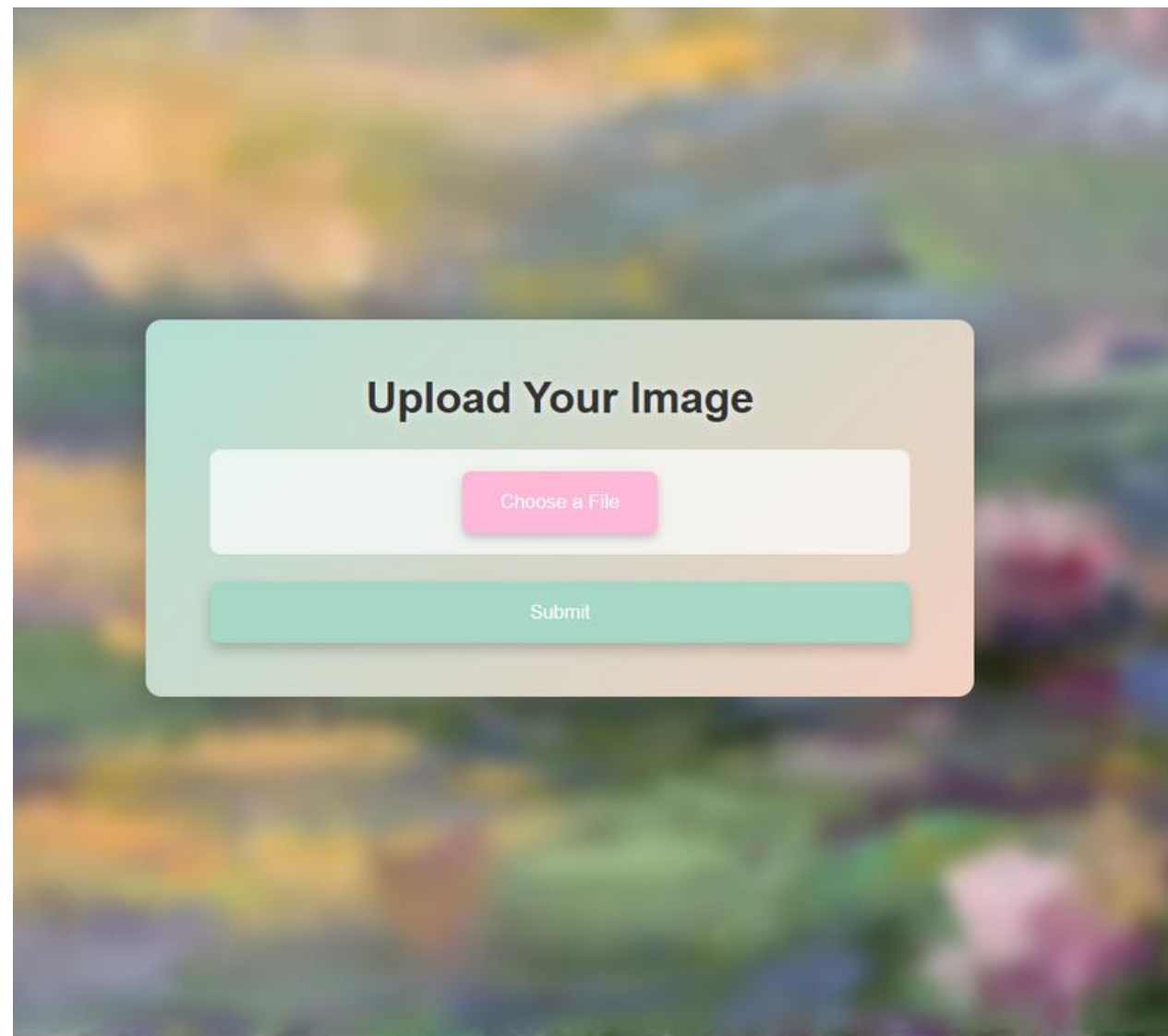
Results



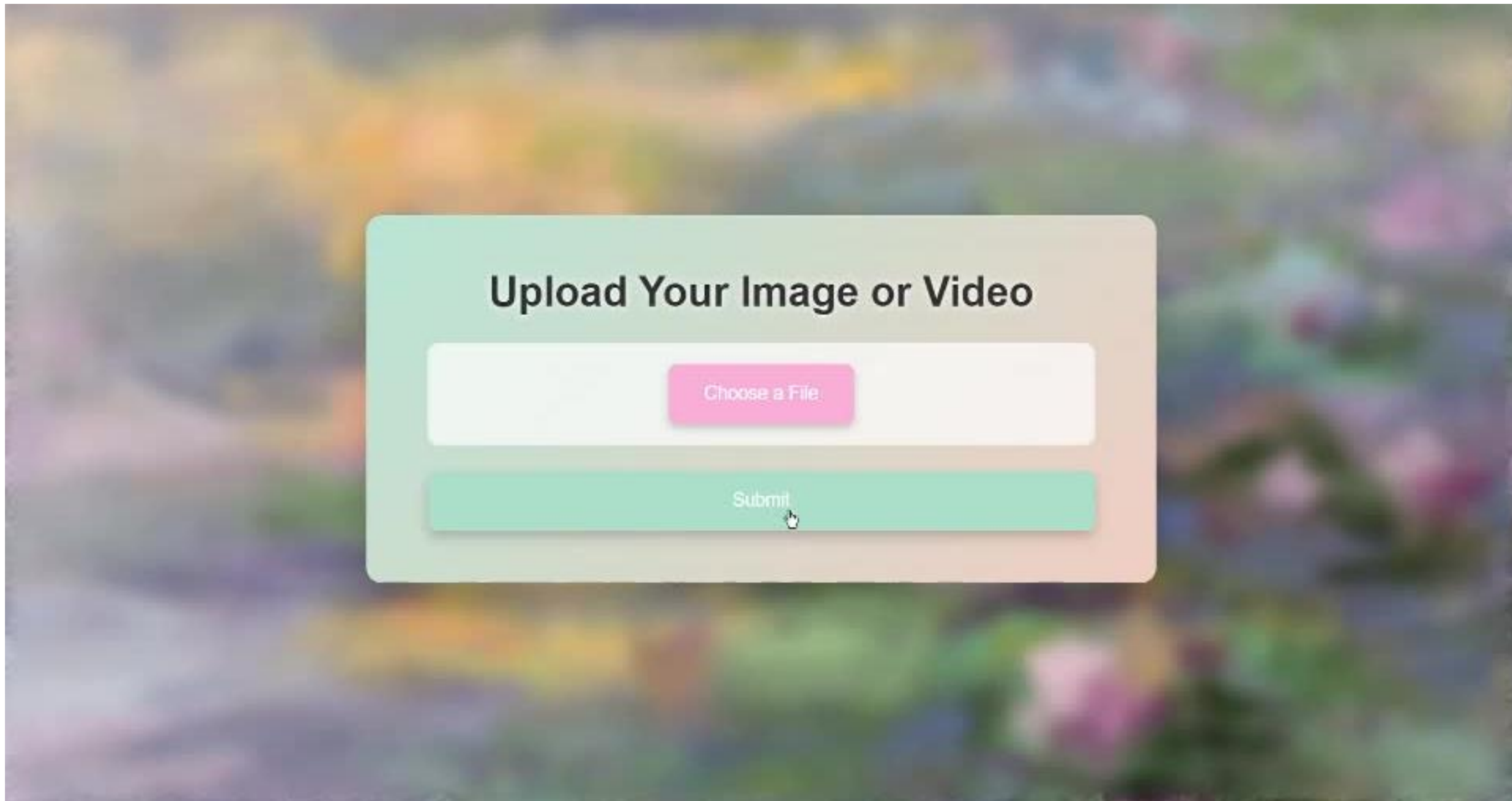


Deployment

Deployment



Deployment



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Conclusion

- **The GAN performed very well in landscape frames which are similar to Monet paintings.**
- **On the other hand, frames with new objects/nature aren't that good.**
- **Also the model lack the variety of styles and the option to choose the desired style.**

Future improvements

- **Train the model on more and more data (more resources).**
- **Add variety of styles and frames.**
- **More Architectures could be tried.**



Questions?

Thanks Samsung
More than family 

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Enabling People

Education for Future Generations

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